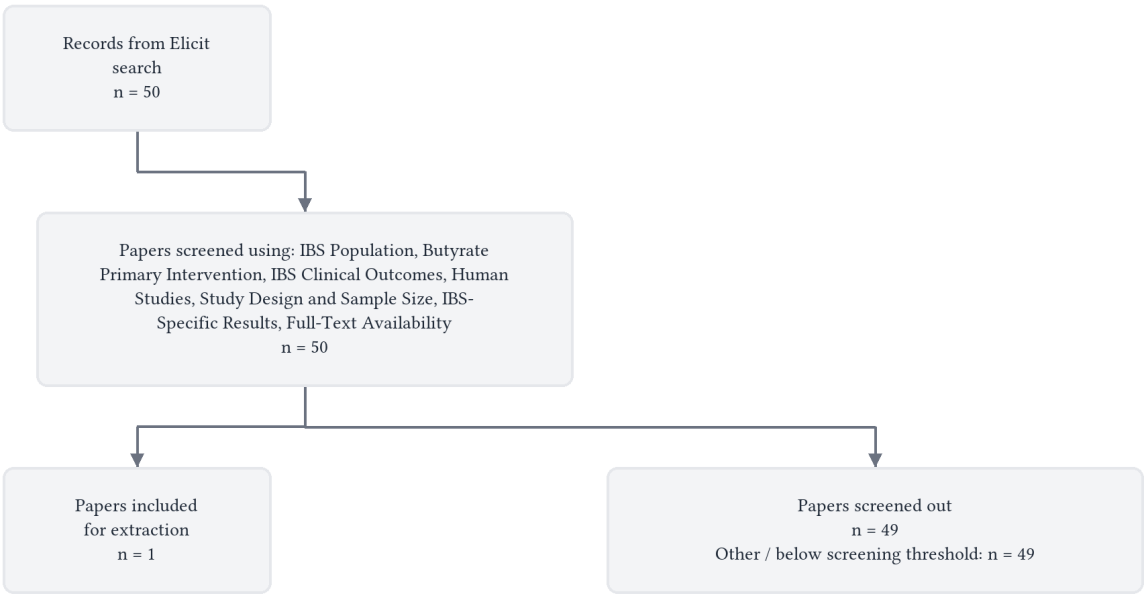
utilidad del butirato en el sindrome de intestino irritable

The available evidence from a single non-randomized study of nearly 3000 patients suggests sodium butyrate may be useful for reducing IBS symptoms and improving quality of life, though controlled trials are needed to confirm its efficacy.

Abstract

A single prospective multicenter clinical trial examined microencapsulated sodium butyrate (150 mg twice daily for 12 weeks) in 2990 IBS patients diagnosed by Rome IV criteria [1, 1]. The study reported statistically significant improvements in all measured gastrointestinal symptoms, including abdominal pain, flatulence, diarrhea, constipation, and urgent bowel movements (all $p < 0.001$) [1]. Quality of life measures improved substantially across all domains, with reductions of 2.49 to 2.95 points on a 0-10 scale representing approximately 40-45% decreases in symptom burden [1]. Patient acceptance was high, with 93.90% willing to continue therapy and 88.90% willing to recommend it to others [1]. However, the study lacked randomization and a control group [1], preventing definitive causal attribution of improvements to sodium butyrate rather than to placebo effects or natural disease variability. While the authors concluded that sodium butyrate may be effective for relieving IBS symptoms through intestinal microbiota modulation [1], the non-randomized design limits the strength of evidence supporting this conclusion. Safety data were limited, though the 6.1% discontinuation rate suggests acceptable tolerability [1].

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "utilidad del butirato en el sindrome de intestino irritable"

The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **IBS Population**: Does the study include participants with irritable bowel syndrome (IBS) diagnosed according to established criteria (Rome criteria, Manning criteria, or physician diagnosis based on clinical symptoms)?
- **Butyrate Primary Intervention**: Does the study investigate butyrate in any form (sodium butyrate, calcium butyrate, butyric acid, microencapsulated butyrate, or butyrate-producing interventions) as the main therapeutic intervention?
- **IBS Clinical Outcomes**: Does the study report at least one clinically relevant IBS outcome such as symptom severity, abdominal pain, bowel movement frequency, stool consistency, quality of life, or adverse events?
- **Human Studies**: Is the study conducted in human participants (not animal or in vitro studies)?
- **Study Design and Sample Size**: Is the study an interventional or observational study (randomized controlled trial, non-randomized controlled trial, cohort study, case-control study, or case series) with at least 5 participants?
- **IBS-Specific Results**: If the study includes participants with mixed gastrointestinal conditions, are IBS-specific results separately reported? (Answer "Yes" if the study includes only IBS participants)
- **Full-Text Availability**: Is the study available as a full-text publication with sufficient methodological detail for quality assessment (not just a conference abstract)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

At abstract screening, the number of papers excluded for each primary reason was:

- **Other / below screening threshold**: n = 49

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Butyrate Preparation**:

  Extract complete details about the butyrate preparation tested for IBS, including:

- Chemical form (sodium butyrate, butyric acid, other salts)
- Formulation type (microencapsulated, triglyceride matrix, enteric-coated, etc.)
- Dosage amount and frequency
- Duration of treatment
- Route of administration
- Any special delivery characteristics

- **IBS Patient Population**:

  Extract characteristics of the IBS patient population studied, including:

- Sample size
- IBS subtype (IBS-D, IBS-C, IBS-M, unspecified)
- IBS diagnostic criteria used
- Disease severity/duration
- Key demographic characteristics (age range, gender distribution)
- Inclusion/exclusion criteria relevant to IBS

- **Study Design Quality**:

  Extract study design and quality indicators, including:

- Study type (RCT, open-label, observational, etc.)
- Blinding status
- Control/comparison group details
- Randomization method if applicable
- Primary outcome definition
- Key methodological strengths or limitations

- **IBS Outcomes Measured**:

  Extract all IBS-related outcomes and measurement methods used to assess butyrate utility, including:

- Symptom scales or questionnaires used
- Specific IBS symptoms measured (abdominal pain, bloating, bowel habits, etc.)
- Quality of life measures
- Biomarkers or objective measures
- Assessment timepoints
- How outcomes were quantified

- **Effectiveness Results**:

  Extract results showing butyrate's effectiveness for IBS outcomes, including:

- Primary endpoint results with statistical significance
- Effect sizes, percentage improvements, or clinical significance measures
- Results for individual IBS symptoms
- Quality of life improvements
- Response rates or proportion of patients improved
- Any dose-response relationships noted

- **Comparison Group Results**:

  Extract details about control/comparison groups and their results, including:

- What butyrate was compared against (placebo, standard care, other treatments)
- Comparative effectiveness results
- Statistical comparisons between groups
- Any head-to-head comparisons with other IBS treatments
- Baseline characteristics comparison between groups

- **Safety and Tolerability**:

  Extract safety and tolerability data for butyrate in IBS patients, including:

- Adverse events frequency and severity
- Treatment discontinuation rates and reasons
- Serious adverse events
- Patient-reported tolerability
- Any contraindications or precautions identified
- Long-term safety data if available

- **Clinical Utility Assessment**:

  Extract authors' conclusions and clinical utility assessments, including:

- Authors' stated conclusions about butyrate utility for IBS
- Recommendations for clinical use
- Patient acceptance or satisfaction measures
- Cost-effectiveness data if mentioned
- Practical considerations for implementation
- Limitations acknowledged by authors

Results

Characteristics of included studies

A single prospective multicenter clinical trial was identified that examined the utility of butyrate for IBS treatment.

Study	Full text retrieved?	Study design	Sample size	Population characteristics	IBS subtypes	Treatment formulation	Duration
K. Lewandowski et al., 2022	Yes	Prospective multicenter clinical trial; non-randomized cross-sectional study [1]	3000 patients enrolled; 2990 complete surveys collected [1]	Mean age 49.22 years; 65% women, 35% men; Rome IV diagnostic criteria [1]	IBS with variable bowel movements (39.29%), IBS-D (29.54%), IBS-C (21.93%), unclassified (9.24%) [1]	Microencapsulated sodium butyrate in triglyceride matrix, 150 mg twice daily, oral administration [1]	12 weeks [1]

The study enrolled adults aged 18 years or older with IBS diagnosed according to Rome IV criteria and in a psychophysical state of health [1]. Exclusion criteria included chronic mental illness, alcohol and/or drug addiction, and contraindications to sodium butyrate [1]. The study design lacked randomization and did not include a control group [1], representing significant methodological limitations despite the large sample size.

Effects of butyrate on IBS symptoms

The study assessed symptom severity and quality of life impacts using patient-reported questionnaires with scales ranging from 0 (no symptoms) to 10 (most severe symptoms) administered before treatment initiation and after 12 weeks of therapy [1].

Gastrointestinal symptom outcomes

Symptom	Statistical significance	Clinical observation
Abdominal pain severity	$p < 0.001$ [1]	Statistically significant improvement observed [1]
Flatulence	$p < 0.001$ [1]	Significant reduction [1]
Diarrhea	$p < 0.001$ [1]	Significant reduction [1]
Constipation	$p < 0.001$ [1]	Significant reduction [1]
Urgent pressure for bowel movements	$p < 0.001$ [1]	Significant reduction [1]
Nausea	$p < 0.001$ [1]	Significant reduction [1]
Vomiting	$p < 0.001$ [1]	Significant reduction [1]

All measured gastrointestinal symptoms demonstrated statistically significant improvements following 12 weeks of sodium butyrate treatment [1]. The consistent p-values below 0.001 across all symptom categories indicate robust statistical significance, though the study did not report specific effect sizes or percentage improvements for individual symptoms [1].

Quality of life outcomes

Quality of life domain	Baseline mean score	Post-treatment mean score	p-value	Absolute improvement
Overall quality of life	6.61 [1]	3.66 [1]	$p < 0.001$ [1]	2.95 points
Social life	6.43 [1]	3.73 [1]	$p < 0.001$ [1]	2.70 points
Professional work	6.08 [1]	3.41 [1]	$p < 0.001$ [1]	2.67 points
Household duties	5.80 [1]	3.31 [1]	$p < 0.001$ [1]	2.49 points
Hobbies	5.62 [1]	3.84 [1]	$p < 0.001$ [1]	1.78 points

Quality of life measures showed substantial improvements across all assessed domains [1]. The magnitude of improvement was largest for overall quality of life (2.95 points on a 0-10 scale), followed by social life (2.70 points) and professional work (2.67 points). All improvements achieved statistical significance at $p < 0.001$ [1]. These improvements indicate clinically meaningful changes, as the reductions represent approximately 40-45% decreases in symptom burden across most quality of life domains.

Patient acceptance and satisfaction

Patient acceptance of sodium butyrate therapy was notably high. Among the 2990 patients who completed the study, 93.90% declared willingness to continue therapy [1] and 88.90% indicated they would recommend sodium

butyrate to other IBS patients [1]. These satisfaction rates suggest strong patient-reported tolerability and perceived effectiveness.

The discontinuation rate was 6.1% due to lack of therapeutic effect [1], indicating that adverse events were not a primary driver of treatment discontinuation. Treatment tolerance was reported as significantly higher compared to previous treatments patients had received [1, 1], though specific adverse event frequencies and severities were not documented [1].

Safety and tolerability

The study provided limited detailed safety data. No specific information was reported regarding adverse event frequencies, severities, or serious adverse events [1]. The authors noted that contraindications to sodium butyrate use existed as part of exclusion criteria but did not specify these contraindications [1]. Long-term safety data beyond the 12-week study period were not available [1].

The low discontinuation rate of 6.1% and high patient willingness to continue therapy (93.90%) indirectly suggest acceptable tolerability [1]. Patient-reported tolerability was described as higher compared to previous treatments [1, 1].

Comparative effectiveness

The study did not employ a placebo control or standard care comparison group [1, 1]. Instead, patients' outcomes with sodium butyrate were compared retrospectively to their previous treatments [1]. Results indicated higher treatment tolerance and effectiveness for sodium butyrate compared to these previous therapies, with estimated lower costs [1, 1]. However, the lack of randomization and prospective control group limits the interpretability of these comparative claims [1, 1].

Clinical utility considerations

The authors concluded that sodium butyrate in a triglyceride matrix may be effective for relieving IBS symptoms through modulation of intestinal microbiota [1]. They recommended its use for reducing gastrointestinal symptoms and improving quality of life in IBS patients [1]. Cost-effectiveness data suggested lower treatment costs compared to previous therapies, though detailed economic analyses were not provided [1].

The formulation's microencapsulation in a triglyceride matrix represents a specific delivery system intended to enhance targeted delivery to the intestinal microenvironment [1]. The dosing regimen of 150 mg twice daily for 12 weeks demonstrated feasibility in a large patient population across multiple centers [1, 1].

The authors acknowledged the limitation of the non-randomized study design [1, 1], which prevents definitive causal attribution of the observed improvements to sodium butyrate rather than to placebo effects, natural disease course variability, or other concurrent interventions.

References

1. Lewandowski K, Kaniewska M, Karłowicz K, et al (2022) The effectiveness of microencapsulated sodium butyrate at reducing symptoms in patients with irritable bowel syndrome. Gastroenterology Review. https://doi.org/10.5114/pg.2021.112681